Stewart, Section 15.7, Exercise 21

Problem

Use a triple integral to find the volume of the given solid.

The solid enclosed by the cylinder $y = x^2$ and the planes $z = 0$ and $y + z = 1$.

Part (a)

Step 1. Name the three boundary surfaces

The word *cylinder* in this section means a surface that is generated by a line that moves parallel to a fixed direction. The surface $y = x^2$ is a *parabolic cylinder*. The generator line is parallel to the z-axis, and the surface projects onto the parabola $y = x^2$ in the xy-plane. The surface does not depend on z, so for every fixed z the cross-section is the same parabola.

The other two surfaces are planes. The plane $z = 0$ is the floor. The plane $y + z = 1$ is the top plane. You can write the top plane as $z = 1 - y$. This form is more useful here, because it gives z as a function of y.

The solid E is the set of points that lie above the floor, below the top plane, and on the concave side of the parabolic cylinder. The symbol E always means this solid, and V always means its volume.

Step 2. Find the shadow of the solid in the xy-plane

The floor $z = 0$ is part of the boundary. Therefore the shadow of E in the xy-plane is exactly the part of the floor that belongs to E. On that floor, the points must lie on the concave side of $y = x^2$, that is $y \geq x^2$, and they must lie below the top plane, that is $z = 1 - y \geq 0$, which means $y \leq 1$.

So the shadow is the plane region

$$D = \{(x, y) : \ -1 \leq x \leq 1, \ x^2 \leq y \leq 1\}.$$

The two surfaces meet where $y = x^2$ and $y = 1$. The solutions are $x = \pm 1$. So the shadow is closed on the left and on the right by the parabola, and on the top by the line $y = 1$. Figure 1 shows the solid in space, and Figure 2 shows the shadow D beside one horizontal slice.

Step 3. Choose the order of integration

Fix a point (x, y) in the shadow D. Draw the line through this point parallel to the z-axis. This line enters the solid at the floor $z = 0$ and leaves the solid at the top plane $z = 1 - y$. Each surface

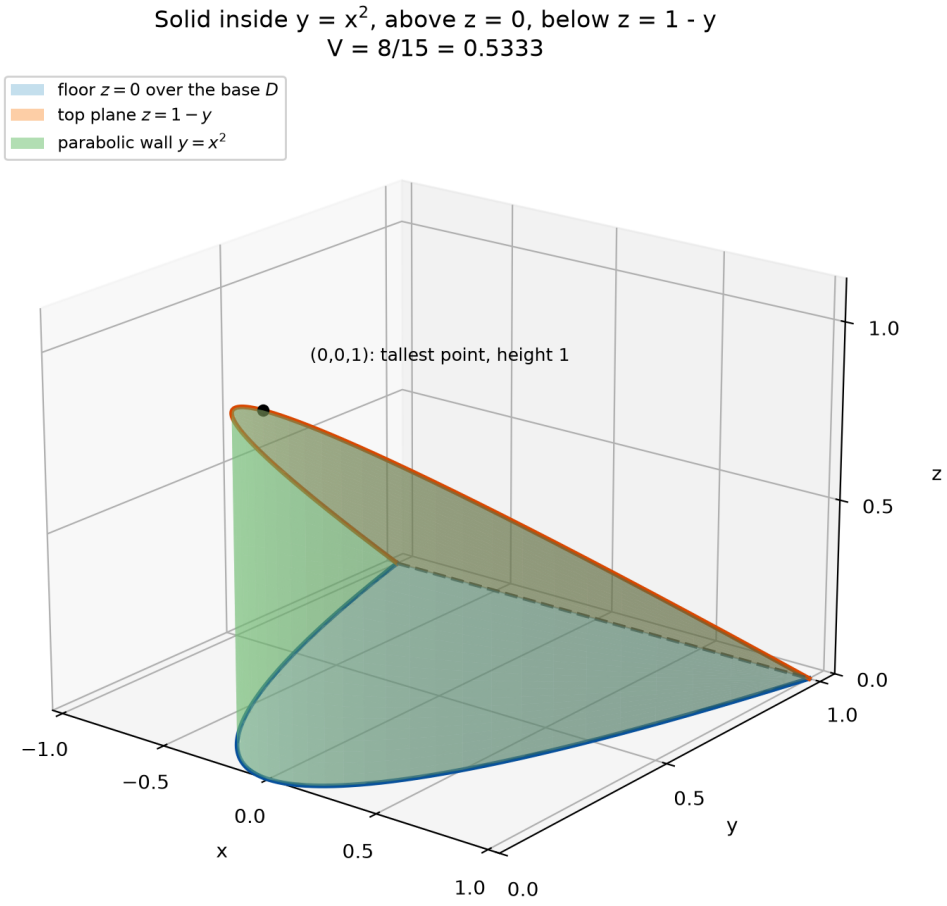


Figure 1: The solid E of Exercise 21. The green surface is the parabolic wall $y = x^2$. The light blue surface is the floor $z = 0$ over the shadow D. The orange surface is the top plane $z = 1 - y$. The blue curve is the parabola $y = x^2$ in the floor, the orange curve is the same parabola lifted to the top plane, and the black point $(0, 0, 1)$ is the tallest point of the solid.

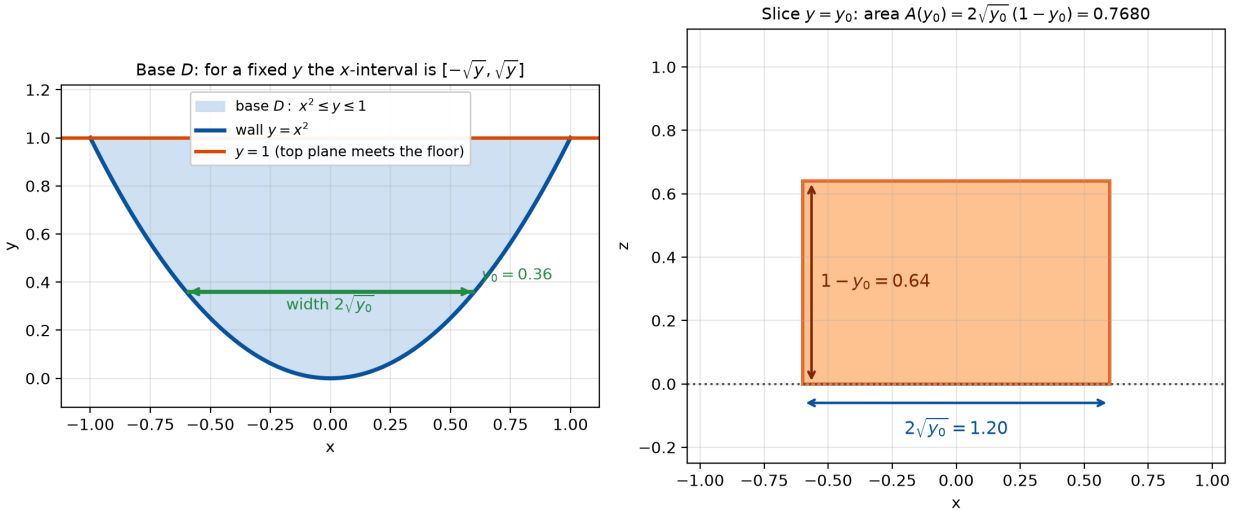


Figure 2: Left: the shadow D in the xy-plane. For a fixed y the x-interval is $[-\sqrt{y}, \sqrt{y}]$, so the horizontal width is $2\sqrt{y}$. Right: the slice of E at the fixed height $y = y_0 = 0.36$. The slice is a rectangle, because at this height z runs from 0 to $1 - y_0$ and x runs from $-\sqrt{y_0}$ to $\sqrt{y_0}$. Its area is $A(y_0) = 2\sqrt{y_0}\,(1 - y_0) = 0.7680$.

gives one value of z for each (x, y), and the two values do not depend on z. So the order $dz\,dy\,dx$ is legal: every vertical line meets E in a single interval of z.

The lower limit is $z = 0$ and the upper limit is $z = 1 - y$. On D we have $y \le 1$, so $1 - y \ge 0$. The upper limit is therefore never below the lower limit. The interval shrinks to a single point only on the line $y = 1$, and a single point adds no volume.

The volume element is $dV = dz\,dy\,dx$. It represents the volume of a small box with sides dx, dy, and dz. A triple integral of 1 over E adds the volumes of all such small boxes, so it gives the volume of E:

$$V = \iiint_E 1\,dV = \iint_D \left(\int_0^{1-y} dz \right) dA = \int_{-1}^{1} \int_{x^2}^{1} \int_0^{1-y} dz\,dy\,dx.$$

Step 4. Integrate in the order z, then y, then x

The inner integral is trivial, because the integrand is 1 and the limits do not contain z:

$$\int_0^{1-y} dz = \Big[z \Big]_0^{1-y} = 1 - y.$$

Now integrate in y:

$$\int_{x^2}^{1} (1 - y)\,dy = \left[y - \frac{y^2}{2} \right]_{x^2}^{1} = \left(1 - \frac{1}{2} \right) - \left(x^2 - \frac{x^4}{2} \right) = \frac{1}{2} - x^2 + \frac{x^4}{2}.$$

Now integrate in x. The integrand is even, so the integral over $[-1, 1]$ is twice the integral over $[0, 1]$:

$$V = 2 \int_0^1 \left(\frac{1}{2} - x^2 + \frac{x^4}{2} \right) dx = 2 \left[\frac{x}{2} - \frac{x^3}{3} + \frac{x^5}{10} \right]_0^1 = 2 \left(\frac{1}{2} - \frac{1}{3} + \frac{1}{10} \right).$$

The bracket has the common denominator 30:

$$\frac{1}{2} - \frac{1}{3} + \frac{1}{10} = \frac{15 - 10 + 3}{30} = \frac{8}{30}.$$

Therefore

$$V = 2 \cdot \frac{8}{30} = \frac{8}{15}.$$

Step 5. Check the value with a second order of integration

A correct answer must not depend on the order of integration. Project E onto the yz-plane. On the projection, $0 \le z \le 1$ and, for a fixed z, the top plane gives $y \le 1 - z$. The condition $y \ge x^2$ says that x runs between $-\sqrt{y}$ and $\sqrt{y}$. So

$$V = \int_0^1 \int_0^{1-z} \int_{-\sqrt{y}}^{\sqrt{y}} dx\, dy\, dz = \int_0^1 \int_0^{1-z} 2\sqrt{y}\, dy\, dz.$$

The inner integral is

$$\int_0^{1-z} 2y^{1/2}\, dy = \frac{4}{3}(1 - z)^{3/2}.$$

Then

$$V = \frac{4}{3} \int_0^1 (1 - z)^{3/2} dz = \frac{4}{3} \cdot \frac{2}{5} = \frac{8}{15}.$$

The two orders give the same value.

Step 6. Check the value with slices and with numbers

A horizontal slice at height y is the rectangle in Figure 2. Its area is

$$A(y) = 2\sqrt{y}\,(1 - y).$$

The slice method gives

$$V = \int_0^1 2\sqrt{y}\,(1 - y)\, dy = 2 \int_0^1 \left(y^{1/2} - y^{3/2}\right) dy = 2 \left(\frac{2}{3} - \frac{2}{5}\right) = 2 \cdot \frac{4}{15} = \frac{8}{15}.$$

The exact value is $8/15 = 0.5333333333\ldots$.

Three independent numerical methods confirm this value. The script `part1_step3.py` computes all three, and every one of them agrees to the precision that its own error bound allows.

Method	Value	Error
Composite Simpson, 2000 panels	0.533333333333	8.3×10^{-15}
Midpoint rule, 2400×2400 grid	0.533318077546	1.5×10^{-5}
Monte Carlo, 4×10^6 points	0.533026500	3.1×10^{-4}

The Monte Carlo value has the statistical error $\pm 1.3 \times 10^{-3}$ at three standard deviations, so it agrees with $8/15$ inside its error bar. As a last sanity check, the solid sits inside the box $2 \times 1 \times 1$, which has volume 2. The value $8/15$ is smaller than 2, as it must be, and it is about 27% of the box, which matches the fraction of the box that the picture shows.

Step 7. State the answer

$$\boxed{V = \frac{8}{15} \approx 0.5333}$$

How to solve problems of this kind

1. **Name each surface.** Decide whether a surface is a plane, a cylinder, a sphere, or a cone. A cylinder in this section is a surface whose generator lines are parallel to one axis. That axis tells you which variable the surface does not contain.
2. **Find the shadow.** Set the two "cap" surfaces equal to each other, or use the floor, to find the plane region D under the solid. Draw D before you integrate. The shadow is the domain of the outer two integrals.
3. **Put the explicit bounds innermost.** The variable that has a lower and an upper bound given by two surfaces should be the inner variable. Here the top plane gives z as a function of y, so z is the inner variable.
4. **Test one line.** Draw a line parallel to the inner axis through a general point of D. If the line enters the solid once and leaves once, the order is legal, and each surface supplies one limit.
5. **Integrate from the inside out.** Do one variable at a time and keep the other variables as constants. Check each antiderivative by differentiating it back.
6. **Check the volume three ways.** Use a second order of integration, use the slice method, and use a numerical method. Three methods that agree are strong evidence that the answer is correct.
7. **Check the size.** Compare the answer with the volume of a box that contains the solid. A volume cannot be negative and cannot be larger than the box.